GridIQ

Site assessment: overview

Demo Corporation · Pretoria Data Center

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment
GQ-A-0001 · 7-day assessment

Report
revision 0 · provisional preview

Observation period (SAST)
2026-11-11 15:00 -> 2026-11-12 07:26

Observed
0.68 days of 7

Site address
12 Server Ave, Pretoria

Measurement points
UPS Input (incomer)

Prepared by GridIQ from measured data. No engineer's sign-off is implied unless a named reviewer appears on this page.

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Executive assessment

Main measurement point: UPS Input. Period 2026-11-11 15:00 to 2026-11-12 07:26 SAST.

The assessment has 16 hour(s) of data so far: too little for a daily pattern. This is a partial picture.

A full day of measurements gives the first operating profile; a full week gives weekday and weekend behaviour and a power-quality screening.

Key numbers

Energy imported
5 176 kWh

Maximum demand
432.6 kVA

Load factor
74 %

Areas to follow up
1 of 4

Energy imported: Register energy over the observed 16 h (register). Maximum demand: Highest 30-minute average (billing basis); contracted 650 kVA. Load factor: Average kW ÷ maximum 30-minute kW. Typical operating load 422.4 kW. Areas to follow up: 1 prioritised observation(s); 2 area(s) with insufficient evidence.

Assessment by area

Area	Status	Basis
Measurement quality	Insufficient evidence	16 h observed so far; at least 24 h are needed for a daily pattern.
Supply observations	Insufficient evidence	Fewer than 24 h of valid voltage values.
Installation and load behaviour	Investigate	load outside operating hours at 100 % of the operating average.
Demand exposure	No material issue observed	Maximum 30-minute demand 432.6 kVA, 67 % of the contracted 650 kVA.

Immediate priorities

- 1. Only 75 % of the expected measurements were received; conclusions for this point are limited. Next: Restore the meter connection; GridIQ will check what data can be recovered. (GridIQ; high confidence)

Important measurement limits

- Observation period 0.7 days (2026-11-11 15:00 to 2026-11-12 07:26 SAST); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.

1. Findings by point

- Investigate (UPS Input, confidence high; for: GridIQ): Only 75 % of the expected measurements were received; conclusions for this point are limited.
- Note (UPS Input, confidence medium; for: site): Outside operating hours the load stays at 100 % of the operating-hours average. If the site is not meant to run around the clock, look for equipment left on.

2. Installation and measurement boundary

Point	Role	Nominal	CT primary	Contracted MDI	Device · installed (SAST)
UPS Input	incomer	230 V	–	650 kVA	device · 2026-08-05 22:45

Point	Coverage	10-min slots measured	Samples	Reporting interval	Gaps >= 30 min
UPS Input	74.6 %	75 of 98	8 767	5 s	1

Coverage: unique 10-minute slots of the observation window; each slot counts min(1, samples × its own reporting interval ÷ 600 s); samples are unique per meter and timestamp (duplicates are rejected at ingest). Missing data is never filled or shown as zero.

sum of 1 top-level incomer(s); sub-meters are inside these and are not added

3. Energy, demand and operating pattern

UPS Input

Energy imported
5 176 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
432,6 kVA at 2026-11-11 22:30

Contracted MDI
650 kVA (67 % used)

Highest instantaneous active power
445,9 kW (minute of 2026-11-11 19:11)

Baseload (10th percentile)
420,1 kW

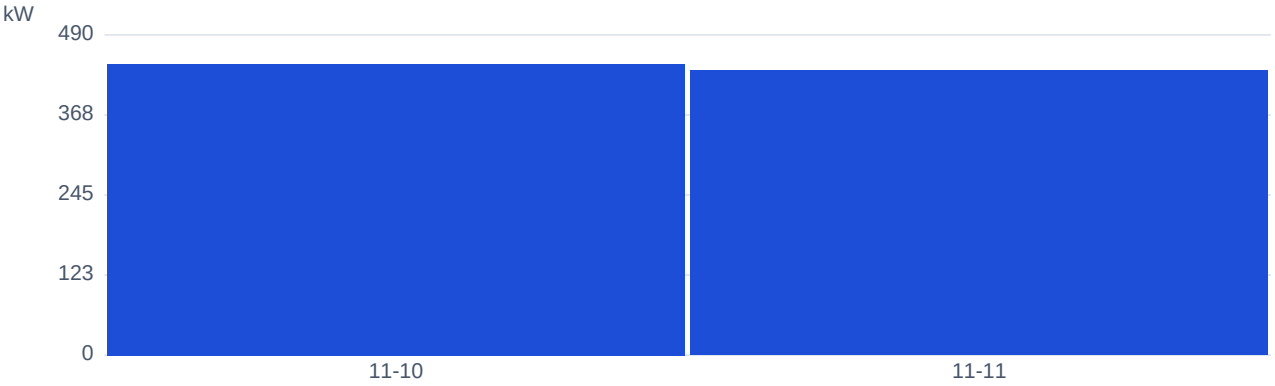
Operating-hours average
422,4 kW

Outside operating hours
424,1 kW

Figure 1. Operating fingerprint: typical day (hourly kW, SAST)

hourly average kW, SAST; Monday to Friday; median with P10–P90 across days when at least 3 days contribute to an hour

Figure 2. Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

4. Electrical summary

- UPS Input: voltage A 201–226,1 V, B 204,4–225,8 V, C 206,3–226,8 V; THD-V p95 A 2.9 %, B 2.8 %, C 2.9 %; PF (loaded) 0.989; 0 minute(s) below 23 V.

Screening against NRS 048-2 thresholds; see the power-quality report for detail and the basis of every figure.

5. Notable events and data limitations

- PROVISIONAL: the assessment is still running; these results cover the elapsed period only and are not final.
- Observation period 0.7 days (2026-11-11 15:00 to 2026-11-12 07:26 SAST); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.

- UPS Input: 75 % of expected measurements received (1 gap(s) of 30 min or more).
- UPS Input: the meter's capability profile is not confirmed; only channels present in the data are reported.

5 voltage/frequency excursion event(s) recorded; inferred from the sampled RMS trend (pq_events), not device-recorded.

6. Prioritised actions

#	Action	Who investigates	Urgency	Confidence
1	Only 75 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high

7. Feasibility-data readiness

Supplied: 30-minute interval load profile (kW, kVA); daily energy; billed maximum demand with timestamps; power-quality screening results; metering topology and observation limits.

Missing inputs: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

GridIQ supplies measured data; system sizing and financial modelling belong to the feasibility study.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
c392b3d15ec80e4a8ac6c3ec0b61508c…

Calculation version
gridiq-calc-2026.11.2

Rule version
gridiq-rules-2026.11.2

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-12 07:26

Generated (SAST)
2026-11-12 07:26

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: c392b3d15ec80e4a8ac6c3ec0b61508c5ab09fe652facac1b7e3e52d570c539a